

Question

Question 1 (10 points)

Which of the following is NOT a characteristic of the scientific method?

Question 2 (10 points)

Question 3 (10 points)

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Question 4 (10 points)

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Question 5 (10 points)

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Question 6 (10 points)

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Question 7 (10 points)

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Tentative Answer / Thesis

Art and science are deeply intertwined. Artistic innovations allow scientists to see, interpret, and communicate phenomena that are otherwise invisible, while new scientific discoveries inspire new artistic techniques. This relationship is co-evolutionary, not one-directional.

1. Making the Invisible Visible

Question: How do artistic techniques reveal phenomena beyond human perception?

Chronological Examples:

- 17th c. – Robert Hooke: Micrographs of cells; artistic drawings emphasize scientifically relevant features.
- 17th c. – Antonie van Leeuwenhoek: Artist-assisted illustrations of microorganisms.
- 16th-18th c. – Stippling: Precision technique adopted from art for scientific accuracy.
- 20th c. – SEM imagery & BioArt: False-coloring and detailed imagery highlight invisible structures.

Pig Atlas Connection:

- Makes embryonic structures visible in ways students can interpret.
 - Shows how visual innovation reveals scientific phenomena.
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2. Educating the Eye

Question: How do artistic techniques train scientists' perception and understanding?

Chronological Examples:

- 16th c. – Vesalius: Anatomical drawings merge aesthetics and empirical observation.
- 18th c. – Hunter, Choulant atlases: Repeated exposure trains perception; standardizes what counts as visible.
- 19th c. – Haeckel illustrations & translucent diagrams: Highlight morphologies and patterns in nature.
- 20th c. – Pig Atlas: Explicitly designed for visual education and standardization.

Pig Atlas Connection:

- Central example of training perception.

- Shows how art structures scientific understanding, making invisible patterns teachable.
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3. Representation as Innovation

Question: How do visual innovations generate new scientific insights?

Chronological Examples:

- 19th c. – Maxwell's thermodynamic surfaces: Abstract relationships made tangible through 3D models.
- Late 19th c. – SEM precursors: Visualization of microscopic structures enhances understanding.
- 20th c. – X-ray imaging: Photographs reveal hidden structures but require interpretive skills.
- 21st c. – NASA imagery, Barabási Lab network diagrams: Data rendered visually creates new conceptual understanding.

Pig Atlas Connection:

- Embryo cross-sections provide early model for representing complex, layered structures.
 - Demonstrates visualization as generative, not merely descriptive.
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4. Reciprocity – Science Inspiring Art

Question: How has scientific advancement inspired new artistic expression?

Chronological Examples:

- 17th-19th c. – Haeckel & natural history illustration: Scientific discoveries shape composition, style, and emphasis in art.
- 20th c. – Photography & mechanized reproduction: Artists adapt to new scientific imaging technologies.
- 21st c. – BioArt, Pixar's scientifically-informed VFX: Scientific discoveries directly inspire new artistic forms and techniques.

Pig Atlas Connection:

- Shows archival inspiration continues: scientific priorities dictate visual emphasis.
 - Highlights co-evolutionary loop: science nominates, art materializes.
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Takeaways

1. Art makes the invisible visible: Hooke → SEM → Pig Atlas → NASA.
2. Art trains scientific perception: Vesalius → Haeckel → Pig Atlas.
3. Visualization drives conceptual innovation: Maxwell → Pig Atlas → Barabási.
4. Science inspires artistic innovation: Haeckel → Pig Atlas → BioArt.
5. Overall: Science nominates what matters; art depicts it; the cycle repeats

Training the Eye is the part of the loop that gives meaning
What is useful and interesting to people varies over time

NEW OUTLINE

- Start with Thesis Question

Slide 1: Title

Visualizing the Invisible

Slide 2:

In this presentation I'm going to attempt to answer the question:

To what extent is advancement in science intertwined with innovations in artistic expression, and how has this relationship evolved to help us visualize the unseen?

Slide 3:

First I'll present to you the answer I came up with and then we'll go through how I got there!

Answer: Through my research I came to the following conclusion:

A co-evolutionary process and feedback loop between nomination, and depiction. Every time this loop begins again, knowledge and meaning are created.

Slide 4: Science:

Lets get into it. We'll start with the role of science, nomination.

IN our class we discussed the transition from natural history to biology. In these discussions we studied the works of Foucault in his Book the Order of Things in which he identified that Natural History's aim was to "nominate the visible". (Include picture of something). Natural History's switch from the classical age's knowledge structure's was focusing on the visible features of the world, a perceptible knowledge. BUt as Natural History turned into our modern conception of science... as natural history bifurcated into history and biology, we became more concerned with the.... Inner workings of animals, why do we categorize and group things the way we do and is there more than the visual with which we can base our classification and understanding on. The further we stray from natural history into the present, the more concerned the study of science becomes with the INVISIBLE

- Microscopes allow us to see things unbenknownest to the naked eye
- We beging to studying the genetic code rather than observing heritable traits
- DNA sequencing
- Microscopy
- Fluorescent tagging

While these are all things that made previously invisble phenomena visible to scientists, older studies of anatomy dealt with dissection, making a part of the body the general public knew existed visible to scientists.

SO we are not only concerned with what is invisible or visible, but to who.

This is my lead in to my first archival example of the loop I mentioned previously

Pig Atlas

As part of our class these past few weeks we had access to the MBL Archives. As I was digging through these archives I came across the following object

Show Pictures= at first glance I thought this was a sculpture, a trinket or sculpture piece belonging to an MBL Researcher, Trinkaus, from the outside I thought it could have been a marine creature sculpted out of paper, but flipping the page I saw that it was a flippable instructional material of a 10 millimeter pig. Blown to the scale of the apparatus, the exterior of each page carved to match the image on the top of each page, the cross section of a pig fetus

- My foremost interest in this came from it's artfulness, the sculpture, but i was then interested in its use as an instructional material
- The development of this tool, likely a tedious timely process for the better of instructing students.
- EUDCATION
 - The instructional nature of this material, of this sculpture is key to its creation,
 - Created by the department of visual education in 1967

- "Dedicated to H. Hudnall Ware Jr, physician administrator educator and humanitarian with the hope that it will help to speed the training of others who will join him in working toward the total betterment of our society"
- This is a printed copy of a reconstruction made several years ago, was printed bc of its popularity
 - It's popularity, its physical printed form, it's visibility gave it meaning and purpose
- ART
 - Quote about the artist: "It is almost fitting that I acknowledge my indebtedness to Mr. John Conway, artist, who prepared all the layout and registrations for this volume. Without his patience, dedication and skill it could not have been completed"
 - 16x magnification of the pig, they are not looking at things at the scale of the pig
 - Not fully realistic, effective communication is not always truthful in its visual representation
 - Really helping students to visualize this, and also wants to show them that this is art (not to scale)
- SCIENCE
 - So what made this possible?
 - Well these crosssections were created because of the BLock Surface Method created by HEegre
 - 1947 BLOCK SURFACE METHOD of staining applied to embryology study
 - In his paper on this method hergre says sutdenst are "handicapped" in thier study of ebyrology by their inability to establish the orientation of serailly sectioned organic material
 - <https://anatomypubs.onlinelibrary.wiley.com/doi/abs/10.1002/ar.1090970103#:~:text=Page%201,orientation%20of%20the%20prepared%20sections>.
 - Allowin them to look at the innards of a pig at 545 levels at intervals of 16 micra
 - Genre was a professor of anatomy ath the medical college of virginia. As a scientist and profssisonal studying anatomy while Hegre may have access to his own specimens and be deeply familiar with the inner workings of a pig, something invisible to most students in their classroom or the public when buying pig at the store... these educational drawn corssections of the pig was modeled into an interactive labeled, redrward carved and printed sculpture with the aim of "trainign others to help better society"

The Loop for the Atlas: we worked backwards but

Science: 1947, the block surface staining method was created to make images of the cross sections of embryonic organisms - this is the making of the visible, and the nominations as shown in these naming of parts in the diagrams.

- Defingin what it is we should aim or try to look at

art : 1967 The sculpture, this educational material was then created to "train the eye", the sculpture makes this 10-13 millimeter pig embryo's insides visible and accessible for students

- Training the eye is an idea that Daston and Galison actually use to talk about their piece of Atlas's of which, this work was called one
 - The atlas aims to make nature safe for science; to replace raw experience-the accidental, contingent experience of specific individual objects - with digested expereince. " pg 85
 - "Atlases supply working objects to the sciences of th eye."

- "Atlases drill the eye of the beginner and refresh the eye of the old hand"
- "Whatever the amount and avowed function of the text in an atlas, which varies from long and essential to nonexistent and despised, the illustrations command center stage... they are the *raison d'être* of the atlases"
- "Eighteenth century atlases required more than mere accuracy"
- artistic aids could "serve as a new language that speak with strength and clearness where written or spoken words could convey their meaning slowly and imperfectly"
- While: "Artists, even slavishly realistic ones, agreed that the artists' very presence meant that images were mediated. "
 - It seems like this MEDIATION was sought after
- In fact: "Burden of representation fell on the audience, instead of on the picture itself" (this begs education i feel like)
- "Pictures became more than merely helpful tools, they were the words of nature itself"
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- "There were many cases where the photograph was unable to provide the audience with a guide equal to that offered by an illustrator.
-

Through this collaboration we get meaning and knowledge

(meaning) This collaboration was specifically stated to aid in the betterment of society and to train scientists and advance the study of embryology beyond the current scientist in the field in 1967

- Klein on Paper Tools:
- Paper tools are material devices in the broader sense of being visible and maneuverable entities that are exterior to mental processes but help to generate mental processes.

OVERALL PIG ATLAS

- Science defined what we should aim to look at, is there something we can't see right now that is worth naming, visualizing, or just simply pointing too
- Art made that visualization possible, accessible, depicted that newfound importance
- And through that, the eye, and the knowledge of scientists, students, and the public is trained, educated, exercised, accustomed etc
- And this meaning, led to more reproductions of this atlas in new forms

Now that we've gone through this full cycle, I'll go through two more examples a bit quicker

Another Archival Material I came across was Haeckel's illustrations

- I was especially drawn to the translucent paper that demarcated which figure on the page to look at with descriptions on the left side of the page

- Talking with the archivist, these translucent pages were meant to keep the artwork intact while still allowing for informational material to be attached to the images
 - Not only were these illustrations aesthetically pleasing, they were instructional and scientific
 - Similar to the Glass Flowers exhibit at the Harvard Natural History Museum
 - Using glass sculpture and aesthetic beauty combined with practicality - scientifically accurate (the particular specimens)
- Haeckel Background:
 - 1834-1919
 - Champion of Darwin's theory of evolution
 - More people learning about Darwin's evolutionary theory than from Darwin's own writings (Robert Richards)
 - Some people called him the "German Darwin"
 - Haeckel coined things like Metazoa, phylogeny, ontogeny, monophyletic, polyphyletic, and "ecology"
 - Visualization of Darwin's theory, - designed phylogenetic trees reflecting evolutionary pathways of various groups of organisms
 - Haeckel's monograph: Radiolaria
 - Single celled aquatic animals
 - <https://www.sciencedirect.com/science/article/pii/S0960982219314368>
 - Colorful lithographs
 - Coast of Heligoland archipelago
 - Napoli drawing
 - Radiolarians
 - Was asked to document newly uncovered species during 19th century
 - He definitely mediated his art through an eye of patterns, symmetry, he was using artistic expression to create a message and intuition, rather than just recordings
 - Based on skeletal geometry

Science -

- Zoologist, discoverer, researcher, discovered new forms, grouped and classified them by morphology and appearance and symmetry and shape, detailed drawings, evolutionary theory, ecology etc
- Degree in comparative anatomy
- Studied organisms in Messina beneath a microscope and was struck by sea creature beauty and geometry and began to sketch
 - Identified them as radiolaria
 - Identified over 100 new species

Art

- Illustrated them artfully - artforms in nature
 - Inspired the art nouveau movement
- Aesthetic style adopted art nouveau

Art forms in nature inspired popularity of the study of natural history

- Promoted idea that science and art could be complementary
- Sketches became central to evolutionary theory

- In 1864, Haeckel sent Darwin copies of his white-on-black Radiolaria illustrations, and over a 20-year period, the men exchanged over 150 letters of correspondence. In 1873, Darwin wrote to Haeckel stating, 'You will do a wonderful amount of good spreading the doctrine of Evolution, supporting it as you do, by so many original observations.' The
- <https://www.artmeetsscience.co/art-forms-in-nature-ernst-haeckel/?srsId=AfmBOorAUUpHgv4z9w7LQTW1xCf1KoZ0GFHusDErfDpV2laan82MFHyv8>
- This art gave scientists a new base to go off of
 - Leading to new scientific classifications of Haeckel's discoveries, discourse around the validity of his taxonomy and groupings, adding to the iterative and corrective process that science takes in its progression
 - Similarly, the geneticist Richard Goldschmidt, stated "... Haeckel's radiolaria were too perfect all over. One had the impression that he first made a sketch from nature and then drew an ideal picture as he saw it in his mind." (Goldschmidt, 1956). With the advent of scanning electron microscopy, some features of radiolarians which one might suspect were invented, were revealed to exist just as Haeckel depicted them (e.g. Emmer, 2007). Note that some of Haeckel's slides are held in the Natural History Museum (London) and have been carefully catalogued for the designation of neotypes (Aita et al., 2009). Another odd opinion made is one with regard to the colors in *Kunstformen* plates. Frank Egerton, a historian of ecology, stated, in the legend of a figure showing plate 11 of *Kunstformen* (spumellarian radiolaria), that "Lithographer Adolph Giltch transformed Haeckel's sketches into forms with somewhat exaggerated colors" (Egerton, 2013). The criticism is remarkable as the forms depicted in the plate have, for the most part, never been seen by anyone other than Haeckel! [https://www.sciencedirect.com/science/article/pii/S1434461024000610#:~:text=Similarly%2C%20the%20geneticist%20Richard%20Goldschmidt,%E2%80%9D%20\(Egerton%2C%202013\).](https://www.sciencedirect.com/science/article/pii/S1434461024000610#:~:text=Similarly%2C%20the%20geneticist%20Richard%20Goldschmidt,%E2%80%9D%20(Egerton%2C%202013).)

Initially, in 1887 Haeckel reported higher taxa of Radiolaria from sediment samples taken during expeditions around the Pacific Ocean, Atlantic Ocean, Arctic Ocean, and Indian Ocean [6,12]. Recently, our knowledge of many species has been revised on the basis of morphological and molecular variation, causing changes in the classification of Radiolarians. These revisions in the group of Radiolarians incorporate recent advances and provide new insight into the taxonomic group of Radiolarians that contradicts the older artificial Haeckel 1887 classification [9,13]. The relatively limited and incomplete classification of these

planktonic species offers substantial opportunity to conduct more taxonomic and systematic research, as exemplified by some recent evolutionary studies [9].
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8002099/>

His work popularized radiolarians among scientist and a general audience

- Again, using art as effective communication

Based classification of radiolaria on their complex patterns of symmetry his groupings on each plate is intentional

Proponent of differentiation and adaptation, evolution, a series of similarities was descent in evolutionary lines

Olaf Breidbach on the art forms in nature forward Offered the first concise outline of a biology that incorporate the principles of evolution as outlined by Darwin

Essentially his works were seen as proof of validity of Darwinism

Displays a succession of symmetries and complexities that show evolution at work

- Evidence of Darwinism although not the mechanism of it

It was a bit stylistic

People describe Haeckel's plates of radiolarians as looking strangely familiar, although never before seen, Haeckel wanted to show that these were forms that were taken from nature, that were natural, so this evocation of familiarity with these forms was not coincidence

- The horns of an antelope look similar to snails and ammonites
- Accentuate the details of each form with distinct color

Plate 65, he draws attention to FORM

- A familiar leaf structure in context with other red algae on the page looks more feather like
- Plays with color too to mediate the image with his findings without words
- Variations in summative three columns looking across you see similarities between the two, the middle being the most symmetrical

Advancing evolutionary theory through the depiction of symmetry and variations in complexity of natural forms

Claims that these are not depictions of existing knowledge but acquisition of knowledge of nature

Revealing more forms of nature knowledge of nature may be ascertained

“Knowledge of nature is natural aesthetics”

The beauty in the microscopic creates, found in the smallest facets of nature reveal to him something about the spirit of human beings

The magic of organic symmetry - the beauty of ornament

Haeckel sees nothing new nothing foreign to nature

- Again directing our eye at what is visible to him, using art to display that, mediating the image

His Style - aestheticized that which appeared strange by making it ornamental

- This is why he refrains from drawing some organism in full - disregards the soft body and looks only at the shell maybe

His approach offered understanding amongst laymen tbc to see nature was for him already knowledge

Bringing scientific illustrations and beauty into art.

- bio

- Finally for my last example, my entry point will start from an artist's depiction
- In 1856 William Henry Perkins accidentally discover mauvine, the first commercialized synthetic dye (aniline purple)
- Purple dye was rare and expensive at the time so he opened a factory that would mass produce this dye
 - This discovery was an aesthetic one, the discovery of this purple dye immediately, became used in the fashion industry, and an aesthetic craze called mauve madness even touched Queen Victoria
 - Perkins was also a collaborator with many painters, was interested in the chemistry of paint

- Science

- Bc of its mass production and the craze around being able to create synthetic dyes to create vibrant colors, fashion, silk dying etc
- Aniline dyes were now deemed possible
- German scientist Paul Ehrlich discovered that it could *stain biological tissues and*
 - *Methylene blue*
 - *Differential staining created*
- *This practice, using dyes to make invisible structures in microscopy visible, is widely used today*
- *8 years after Perkins discovery Gerlach in 1863 used a natural dye to stain brain tissue*
 - *So we knew that coloring tissues could unlock visibility at a different scale*
 - *Synthetic dye use in histology became commonplace by Heinrich Waldeyer*
 - *Then microbiology - Robert Koch used methylene blue and other aniline dyes to stain bacteria*
 - *Then Paul Ehrlich differential staining*

- *Then the Gram stain!*

